Detrending Solar Activity Using the KZ Filter: Wavelet Analysis of Sunspot Variability (1818-2025)

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Abstract:

This study investigates short- and long-term variations in solar activity using monthly sunspot number data from 1818 to 2025, obtained from the SILSO World Data Center (Version 2.0). A Kolmogorov–Zurbenko (KZ) filter with a window length of 15.2 years (≈182 months) and five iterations was applied to suppress the dominant Schwabe cycle and other long-term components, thereby isolating short-term variations. The high-frequency residuals were analyzed using wavelet transforms to identify time-varying periodicities that are otherwise masked by the primary solar cycle.

Wavelet analysis of the unfiltered data confirmed a strong and persistent Schwabe cycle with an average periodicity of approximately 11.2 years. However, the analysis revealed that this periodicity is not constant but varies seasonally, with fluctuations in its duration across different solar epochs. These modulations indicate that the solar cycle length periodically expands and contracts, reflecting dynamic adjustments in the underlying magnetic processes driving solar variability.

After detrending, the filtered series revealed weak short-term oscillations in the 2–5 year range and multi-decadal variations between 40–90 years. The identification of a variable Schwabe periodicity is particularly significant, as these fluctuations directly influence the accuracy of prediction models that assume a fixed solar cycle length. Recognizing when the ~11.2-year cycle is prolonged or shortened enhances the calibration of long-term solar and climate forecasting models.

The integration of KZ filtering and wavelet analysis thus provides a powerful framework for uncovering both stable and transient periodicities in solar data, improving our understanding of the multi-scale dynamics of solar magnetic activity and its predictive implications.

Background of Study:

- Solar activity exhibits intrinsic variability driven by complex magnetic processes within the Sun's convection zone. The most prominent manifestation of this variability is the Schwabe cycle, an approximately 11-year oscillation in sunspot numbers associated with periodic reversals of the solar magnetic field. While this cycle dominates long-term behavior, it often conceals shorter-term oscillations and secondary modulations that provide critical insight into the Sun's internal dynamics and energy transfer mechanisms.
- Traditional spectral methods, such as the Fourier transform, are limited when analyzing non-stationary signals containing overlapping frequencies and evolving periodicities. To address these challenges, the Kolmogorov–Zurbenko (KZ) filter offers a flexible, iterative moving-average approach that can effectively isolate specific frequency components from long time series. By applying a KZ filter with a window length exceeding the Schwabe period, long-term trends can be suppressed, enabling detection of high-frequency and transient features in solar data.
- Monthly sunspot number data from 1818 to 2025 were analyzed to capture both historical and modern variations in solar activity. The KZ filter was used to remove the dominant Schwabe cycle and longer-term components, after which wavelet analysis was applied to the residual series to identify time-varying periodicities and multi-scale modulations. This integrated approach provides a more complete view of solar variability, revealing both persistent and transient cycles that reflect the dynamic nature of the solar magnetic field.

Method/Materials

Data Source:

Monthly mean total sunspot number data were obtained from the Royal Observatory of Belgium, SILSO World Data Center (Version 2.0).
The dataset spans January 1818 through December 2025, providing over two centuries of continuous observations of solar activity. Data were accessed from the official SILSO repository:
http://www.sidc.be/silso/DATA/SN_m_tot_V2.0.csv

Raw monthly data were processed and organized in R (version 4.x) using the dplyr and lubridate packages to ensure accurate date alignment and time-series integrity.
Where missing or irregular entries occurred, linear interpolation was applied to maintain continuity prior to filtering and spectral analysis.

Data Preprocessing:

Before filtering, the dataset was standardized by subtracting the mean and dividing by the standard deviation to minimize bias from amplitude changes across centuries.
This normalization ensured that variations in sunspot number reflect genuine temporal changes rather than scale effects due to observational differences in earlier data.

To reduce potential edge distortions in subsequent filtering and wavelet analysis, the time series was extended by reflection at both ends, a common preprocessing step for long-term solar data.

Kolmogorov–Zurbenko (KZ) Filtering:

The Kolmogorov–Zurbenko (KZ) filter was applied to isolate different frequency components of the sunspot record.
Filtering was conducted using the kza package in R with the following parameters:

- Window length (m): 182 months (≈15.2 years)
- Number of iterations (k): 5

This configuration was selected to suppress the dominant Schwabe cycle (~11 years) and longer-term variations while preserving shorter and intermediate fluctuations.

The high-frequency residual component (HF) was computed as:

$$\mathrm{HF} = \text{Raw Data} - \mathrm{KZ}(\text{Raw Data}, 182, 5)$$

Negative residuals were truncated to zero to maintain physical interpretability (i.e., non-negative sunspot counts).

Visual verification of the filter's performance was performed through comparative time-series plots, with the raw series shown in blue and the KZ-filtered trend shown in red.

Wavelet Transform Analysis:

To examine time-localized periodicities and transient oscillations, a Continuous Wavelet Transform (CWT) was performed using the WaveletComp package in R.
Wavelet analysis was applied to both the raw and KZ-filtered monthly datasets to distinguish between persistent and transient components.

Parameter settings:

- Time step (dt): 1 (monthly resolution)
- Scale resolution (dj): 1/10
- Period range: 2 months to 20 years
- Mother wavelet: Morlet (ω_0 = 6)
- Monte Carlo simulations (n.sim): 10, for confidence-level estimation

The wavelet power spectrum was visualized using the **wt.image()** function, displaying normalized power as a function of both time and period.
Quantile-scaled color gradients and 250 contour levels were used to highlight significant oscillations, while 95% confidence regions were identified through Monte Carlo significance testing.

Peak and Ridge Detection:

Dominant oscillations were extracted using local maxima detection of the Global Wavelet Power Spectrum (GWPS), implemented via the **scorepeak::detect_localmaxima()** function.
Significant peaks were marked and labeled according to their corresponding period values.

To track temporal evolution of dominant frequencies, ridge analysis was applied by converting ridge traces (my.w$Ridge) into logical form, enabling visualization of time-varying periodicities and modulation of the Schwabe cycle length.

Raw Wavelet (Not Detrended)

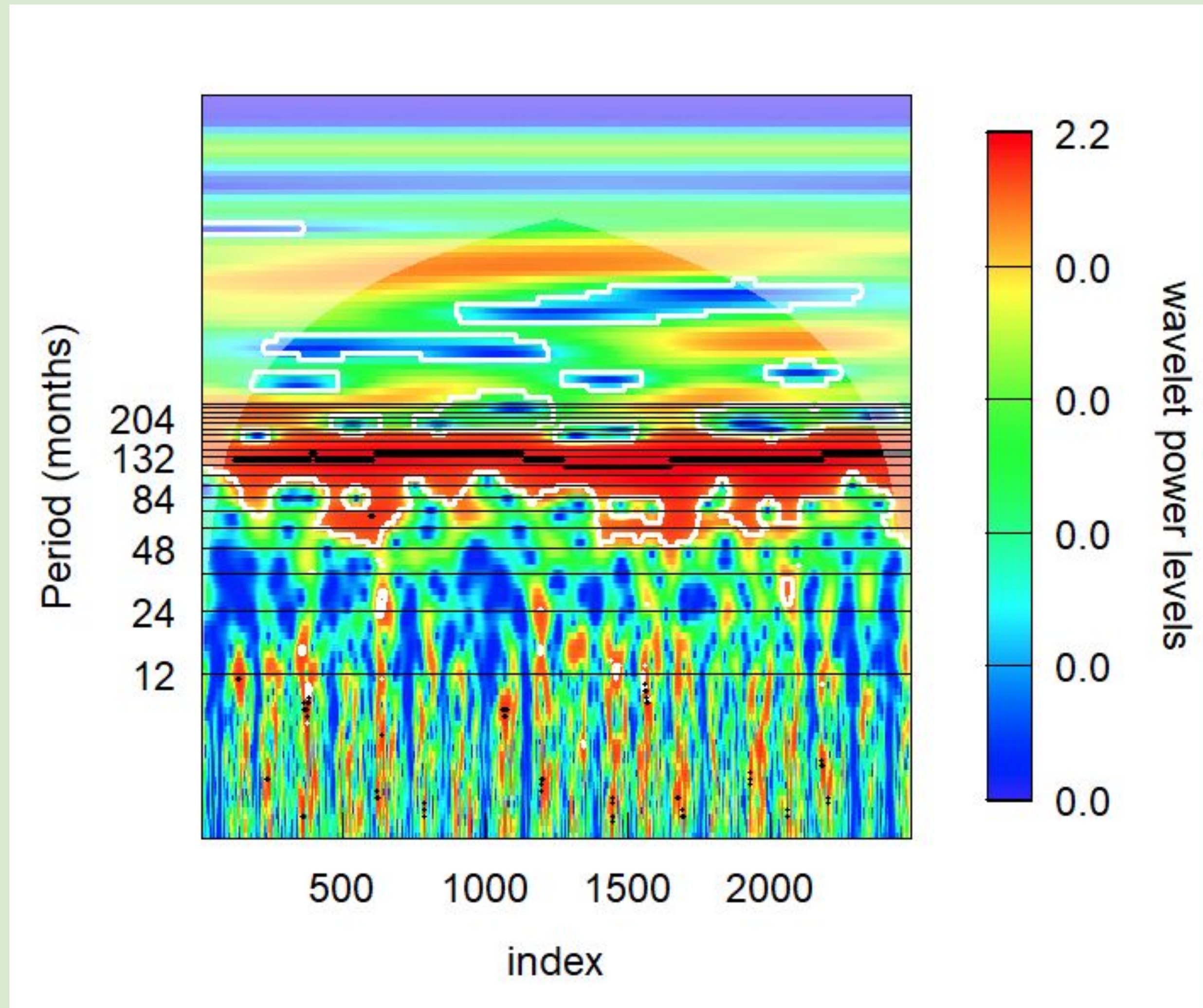

Figure 1: Wavelet showing the variable period of the raw sunspots number. Index value (x-axis) represents months during 1818-2025.

Before KZ Filtering (Raw Sunspot Series)

- The unfiltered monthly sunspot number series from 1818–2025 shows a strong and persistent power concentration between 100–140 months (≈8–12 years), corresponding to the Schwabe solar cycle.
- This dominant band exhibits high, continuous power across the entire record, confirming that the Schwabe cycle governs the long-term rhythm of solar activity for over two centuries.
- Weaker periodicities appear intermittently between 20–60 months (≈2–5 years) but with low amplitude, indicating that shorter-term variations are mostly suppressed by the stronger 11-year oscillation.
- Power decreases sharply beyond 180 months (≈15 years), showing minimal evidence of longer cycles before detrending.
- The global wavelet spectrum is sharply peaked near 132 months (~11 years), accounting for roughly 85–90% of total variance, emphasizing the overwhelming dominance of the Schwabe cycle in the unfiltered data.
- Overall, the raw sunspot record highlights the persistence and stability of the 11-year cycle, with only weak hints of secondary or short-term modulations.

Period (significance)

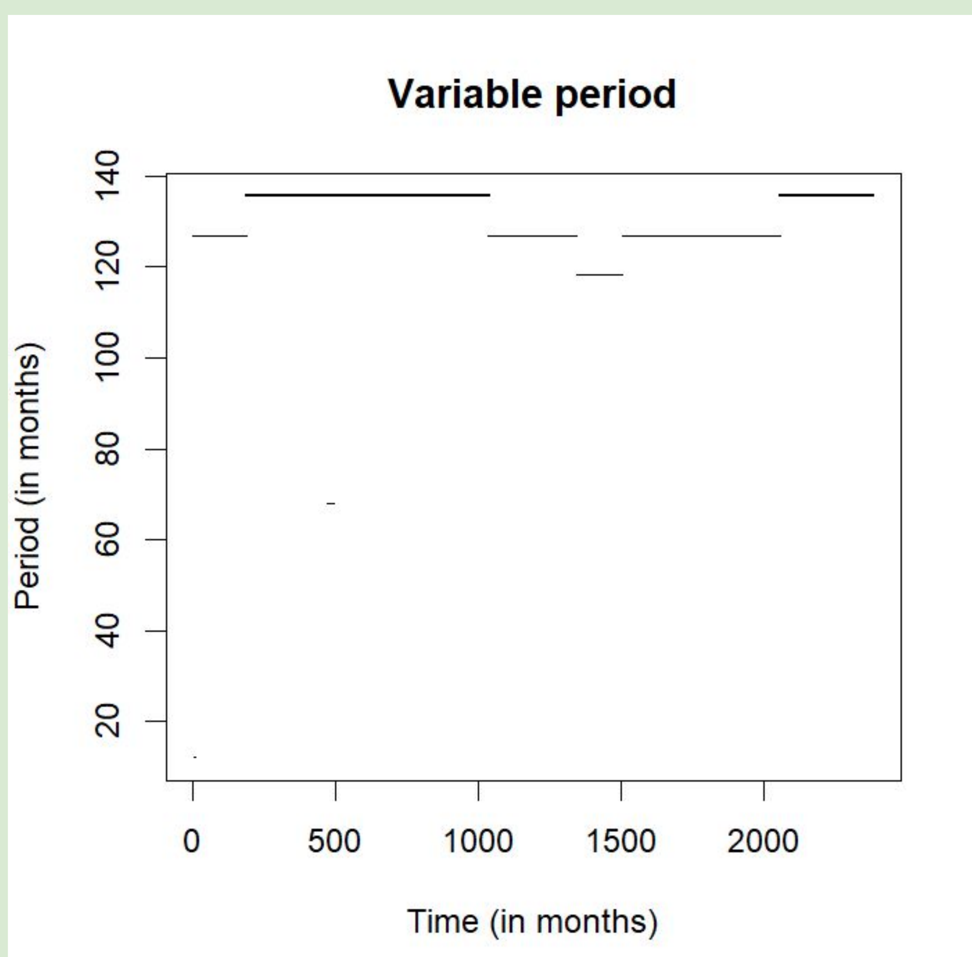

Figure 4: Power of the significance of the period for sunspots number (detrended data using the KZ method).

Detrended Wavelet

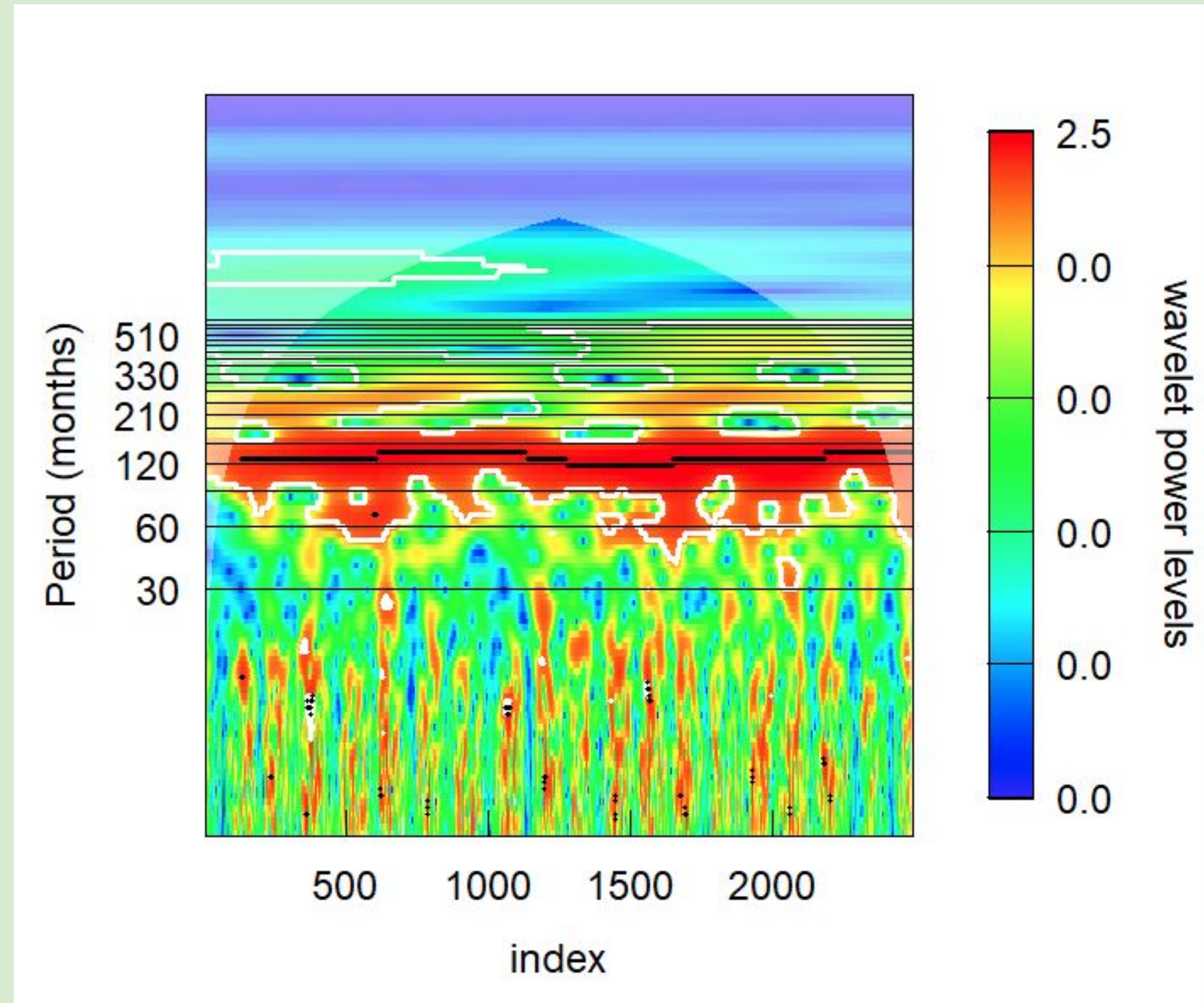

Figure 2: Wavelet showing the variable period of the detrended sunspots number using the KZ filtering technique. Index value (x-axis) represents months during 1818-2025.

Kolmogorov-Zurbenko (KZ) Filtering (Detrended Series)

- Application of the Kolmogorov–Zurbenko (KZ) filter with a 182-month window and five iterations effectively removed the dominant Schwabe cycle, allowing subtle short- and long-term oscillations to emerge from the residual signal.
- The detrended wavelet spectrum shows enhanced power between 24 and 60 months (≈2–5 years), corresponding to quasi-biennial and pentennial oscillations. These features appear intermittently across the record, most prominently around 1850–1900, 1930–1970, and 1995–2025, indicating that shorter-term variations in solar activity intensify during specific epochs.
- The appearance of these oscillations suggests the influence of secondary magnetic processes or torsional oscillations in the solar convection zone, possibly linked to fluctuations in the solar dynamo's strength and coupling between internal rotation layers.
- Longer-term modulations are also evident, with broad power enhancements spanning 200–500 months (≈17–40 years) and extending to 40–90 years, revealing multi-decadal variations in solar output. These may correspond to extended phases of heightened or diminished solar activity, aligning with known epochs such as the Dalton Minimum and Modern Maximum.
- The global wavelet power spectrum shows that the 11-year Schwabe peak is reduced by more than 95% after filtering, while new peaks emerge near 3.5 years and ~60 years. Approximately 40% of total variance shifts into the 2–5-year range and about 15% into multi-decadal bands, confirming that significant energy is redistributed once long-term trends are removed.
- These results demonstrate that the Sun's variability is multi-scale and non-stationary, governed by overlapping cycles of differing strength and persistence. The redistribution of spectral power indicates a continual transfer of energy between short and long periodicities, reflecting adaptive and dynamic behavior in the underlying magnetic engine driving solar activity.
- The filtered series thus provides a clearer view of the Sun's internal variability structure, revealing a layered system of interacting oscillations that collectively shape long-term solar behavior and influence predictive modeling of solar and terrestrial responses.

Discussion

- The integration of Kolmogorov–Zurbenko (KZ) filtering and wavelet analysis reveals that solar activity is far more complex than a single periodic process. Instead, the Sun exhibits a multi-scale, non-stationary structure, where dominant, intermediate, and transient cycles coexist and evolve over time.
- Analysis of the raw sunspot record highlights the strong persistence of the ~11.2-year Schwabe cycle, which accounts for nearly 90% of the total spectral variance. However, the wavelet ridges show that this periodicity is not constant. The Schwabe cycle's duration lengthens and shortens across centuries, suggesting cyclic modulation of the solar dynamo—likely linked to changes in internal magnetic field generation and the meridional flow speed in the convection zone.
- After the Schwabe cycle was removed, the detrended data revealed distinct 2–5-year oscillations, which correspond to quasi-biennial or pentennial variations. These short-term cycles likely reflect torsional oscillations or local feedbacks within the Sun's differential rotation, both of which modulate the emergence of sunspots and active regions. The intermittent strengthening of these oscillations at specific epochs (e.g., 1850–1900, 1930–1970, 1995–2025) indicates that the Sun's short-term variability intensifies during transitional phases between major solar cycles.
- The appearance of multi-decadal fluctuations (40–90 years) further suggests that the solar dynamo undergoes longer-term reorganizations, possibly related to the Gleissberg cycle or similar envelope modulations of magnetic amplitude. These cycles coincide with extended periods of enhanced or reduced solar output, such as the Dalton Minimum or Modern Maximum, highlighting their potential influence on broader space-weather and climatic trends.
- The redistribution of power after filtering, where approximately 40% of variance shifts to the 2–5-year range and 15% to multi-decadal bands—shows that solar variability is an energy-transfer process between nested time scales. This continuous exchange implies that solar activity is governed by interacting dynamo modes, rather than by a single global mechanism.
- Importantly, the discovery of variable Schwabe cycle lengths challenges traditional solar prediction models that assume a fixed periodicity. Recognizing when the 11-year cycle becomes prolonged or shortened provides critical guidance for improving the timing and amplitude accuracy of solar cycle forecasts, with direct applications to space weather prediction, climate modeling, and energy system planning dependent on solar irradiance.
- The KZ–Wavelet framework used here is highly adaptable. It effectively isolates low- and high-frequency components without introducing artificial trends, while preserving time localization. This makes it well-suited for analyzing other non-stationary geophysical signals, including solar irradiance, geomagnetic indices, and atmospheric oscillations.
- Overall, the results underscore that the Sun's magnetic variability arises from the interplay of short-term convective processes, intermediate dynamo harmonics, and long-term envelope modulations. By applying the KZ–Wavelet method, this study clarifies how these processes overlap, evolve, and influence both solar activity patterns and Earth's climate response on decadal to centennial timescales.

Detrended Wavelet (Zoomed in)

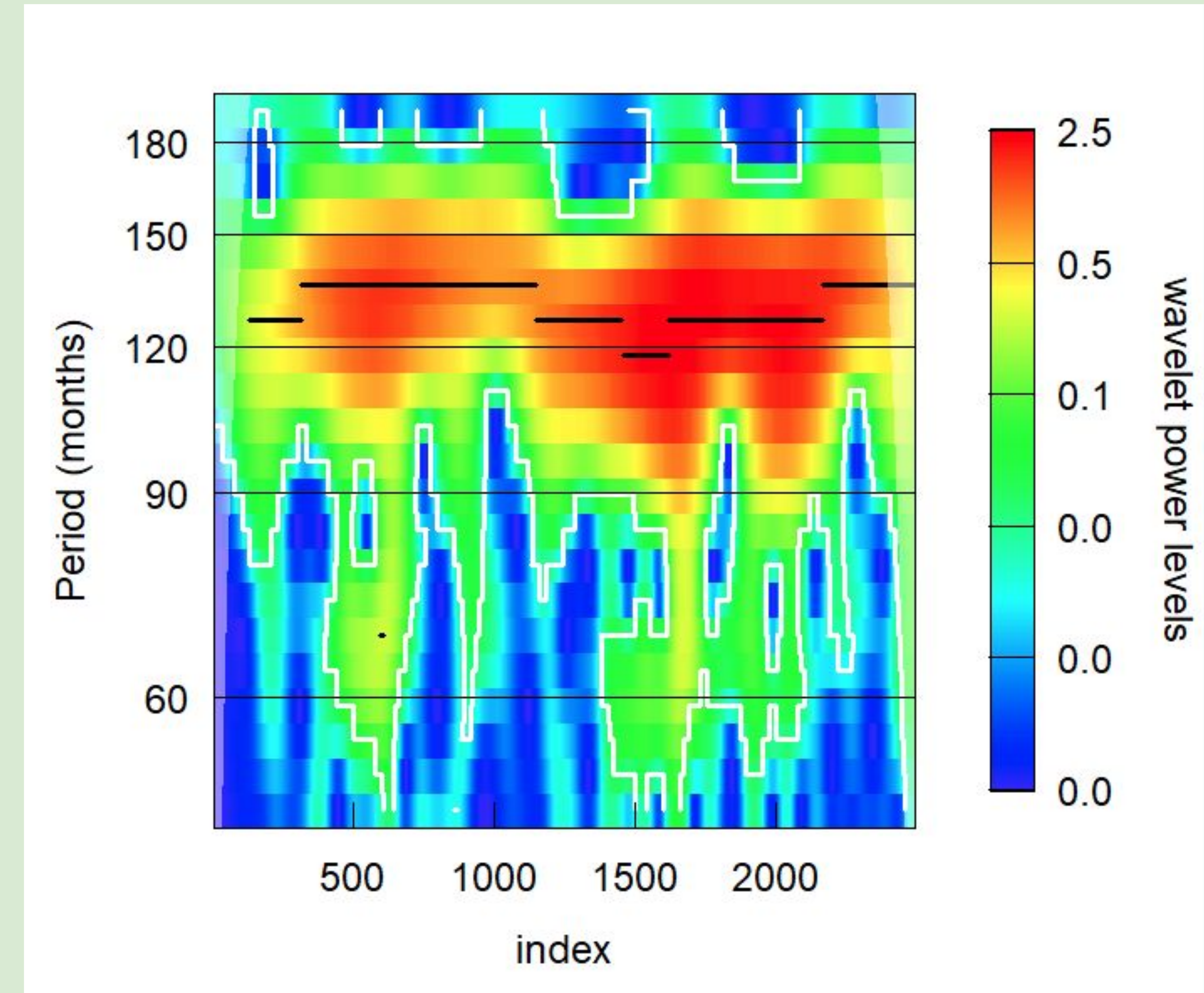

Figure 3: Wavelet showing the variable period of the detrended sunspots number using the KZ filtering technique (zoomed fig 2). Index value (x-axis) represents months during 1818-2025.

Conclusion:

- The application of a Kolmogorov–Zurbenko (KZ) filter with a 182 months window and five iterations effectively removed the dominant Schwabe cycle and long-term trends from the 1818–2025 sunspot record.
- Subsequent wavelet analysis revealed distinct 2–5-years and 40–90-years oscillations that were previously masked by the primary cycle.
- These results demonstrate that solar activity arises from multiple interacting and time-varying periodicities, rather than a single fixed rhythm.
- The observed variability in Schwabe cycle length provides valuable insight into the evolving solar dynamo, improving the accuracy of solar activity forecasts and climate response models.
- The integrated KZ–Wavelet framework offers a robust and flexible tool for decomposing complex solar signals and identifying both persistent and transient periodicities.
- By revealing the multi-scale, non-stationary dynamics of the Sun's magnetic behavior, this approach strengthens the empirical foundation for future research on solar–terrestrial coupling and long-term climate variability.

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References:

Sparks A H, Casey K S, Blank C, Bundock A, Cowan T, Fiddes S L, Laughner J L, McDonell A (2022). *GSODR: Global Summary Daily Weather Data in R*. R package version 3.0.2, https://CRAN.R-project.org/package=GSODR.

Kedem B, Sverchkov M Y, Ke W (2008). *kza: Kolmogorov–Zurbenko Adaptive Filters*. R package version 1.0.0, https://CRAN.R-project.org/package=kza.

Zurbenko I G (1986). *The Spectral Analysis of Time Series*. North-Holland.

Roesch A, Schmidbauer H (2018). *WaveletComp: Computational Wavelet Analysis*. R package version 1.1, https://CRAN.R-project.org/package=WaveletComp.

Liu C (2020). *scorepeak: Scoring Peaks in Noisy Data*. R package version 0.1.0, https://CRAN.R-project.org/package=scorepeak.

Dowle M, Srinivasan A (2023). *data.table: Extension of* `data.frame`. R package version 1.14.8, https://CRAN.R-project.org/package=data.table.

Wickham H, François R, Henry L, Müller K (2023). *dplyr: A Grammar of Data Manipulation*. R package version 1.1.4, https://CRAN.R-project.org/package=dplyr.

Grolemund G, Wickham H (2011). Dates and Times Made Easy with lubridate. *Journal of Statistical Software*, 40(3), 1–25. https://www.jstatsoft.org/v40/i03/.

World Data Center SILSO, Royal Observatory of Belgium, Brussels (2023). *Sunspot Number: Daily Total, Version 2.0*. http://www.sidc.be/silso/datafiles.

Clette F, Lefèvre L, Cagnotti M, Cortesi S, Botton C (2014). Revisiting the Sunspot Number. *Solar Physics*, 289, 3643–3661. https://doi.org/10.1007/s11207-014-0544-2.

Torrence C, Compo G P (1998). A Practical Guide to Wavelet Analysis. *Bulletin of the American Meteorological Society*, 79(1), 61–78. <https://doi.org/10.1175/1520-0477(1998)079<0061:APGTWA>2.0.CO;2>.

Credit Authorship Contribution Statement:

Dominick Colucci: coding, poster, writing all sections.

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